

Dynamics of a Quantum Phase transition in 2D Bose Hubbard Model.

A Dissertation for
PHY-651 - Dissertation

Credits: 16

Submitted in partial fulfillment of Masters Degree
M.Sc in Physics (Computational Physics)

Goa University

In the subject of Physics

by

Apeksha Mahesh Phadte

Seat no: 23PO430005

ABC Id: 454056363905

PRN: 201810848

Under the supervision of

Dr. Bhargav Alvani

School Of Physical and Applied Science
Physics discipline



GOA UNIVERSITY
गोवा विद्यापीठ

Goa University

May 2024



Seal of the School

Examined by:

Declaration

I hereby declare that the project work entitled "*Dynamics of a Quantum Phase Transition in the Two-Dimensional Bose-Hubbard Model*" has been carried out by me under the supervision of Dr. Bhargav Alvani at Goa University.

This report is a bona fide record of original work undertaken by me and has not been submitted previously, either wholly or in part, for the award of any degree or diploma in any other institution.

All sources of information and references have been duly acknowledged

Date: 12/5/25

Place: Goa University

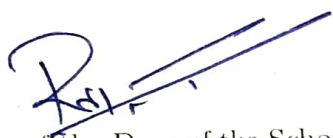


Signature of Candidate

Apeksha Mahesh Phadte

COMPLETION CERTIFICATE

This is to certify that the dissertation report “Dynamics of a Quantum Phase Transition in the Two-Dimensional Bose-Hubbard Model” is a bonafide work carried out by Apeksha Phadte under the supervision of Dr. Bhargav Alavani in partial fulfillment of the requirements for the award of the degree of the Master of Science in Physics at the School of Physical and Applied Science, Goa University.



Signature of the Dean of the School



Seal of the School

Date: 12/5/25

Place: Goa University



Signature of Supervising Teacher

Dr. Bhargav Alavani

PREFACE

This project report presents the study of the dynamics of a quantum phase transition in the two-dimensional Bose-Hubbard model using the mean-field approximation. The work involved developing numerical simulations, analyzing the behavior of the system during a linear quench, and studying critical phenomena like the growth of superfluid domains and freeze-out scaling.

Through this project, I had the opportunity to deepen my understanding of quantum phase transitions, numerical methods in many-body physics, and the subtle role of coherence and fluctuations in out-of-equilibrium systems.

I hope this report will be a valuable reference for future students interested in similar areas of research.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my project supervisor, Dr. Bhargav Alavani, for their continuous guidance, encouragement, and support throughout the course of this project. Their insightful suggestions and critical feedback greatly helped me to complete this work.

I also wish to thank the Department of Physics, Goa University, for providing the necessary resources and a stimulating academic environment.

Finally, I would like to thank my family and friends for their encouragement, patience, and constant motivation during the entire duration of this work.

ABSTRACT

In this project, the dynamics of a quantum phase transition in the two-dimensional Bose-Hubbard model were studied by solving the Hamiltonian of the 2D Bose-Hubbard model using the mean-field approximation. The Hamiltonian was diagonalized and the ground state vectors were achieved. The system was initially prepared in the Mott insulating phase by keeping the initial value of hopping amplitude $J(t)$ around 0.03 and then it was evolved through the critical point into the superfluid phase by applying a linear quench protocol to the hopping amplitude.

Important Quantities such as the average superfluid order parameter over the entire lattice and average the particle density over the entire lattice were calculated throughout the evolution. The phase diagram was first obtained by static mean-field calculations. The order parameter showed the development of superfluid properties through its time-dependent behaviour while the lattice domain size increased to prove superfluidity.

The spatial representation of ψ shows how topological defects occur in these specific snapshots after the quench. When $J(t)$ increases superfluid order parameters appear and expand because this change indicates the formation of extended phase-coherent regions.

The analyses used the overlap function to determine the system's freeze-out time from which the dependency on quench time was measured. The experimental findings demonstrate qualitative match with theoretical predictions using the Kibble-Zurek mechanism through their power-law observation.

The study delivers important understanding regarding non-equilibrium behavior across a continuous phase transition by showing how domains evolve through defects while phase coherence requires time to develop fully.

Contents

1	Introduction	2
1.1	Optical Lattices	2
1.2	Superfluid–Mott Insulator Transition	2
1.3	Literature Review	3
1.4	Dynamics: A Brief Review	5
1.5	Motivation	5
1.6	Aim	6
1.7	Plan of the Report	6
2	Methodology	8
2.1	Mean Field Theory on 2D Bose-Hubbard Model	8
2.2	Dynamics of the Quench Process	11
2.3	Calculation of Freeze-Out Time Using the Overlap Function . . .	13
2.4	Analysis of Domain Size	14
3	Results	15
3.1	Phase transition	15
3.2	Linear Quench Profile: Variation of Hopping Amplitude $J(t)$ with time t	17
3.3	Time Evolution	18
3.4	Domain Size Growth During the Quench	22
3.5	Analysis of Freeze-Out Dynamics Across the Quench	23

Chapter 1

Introduction

1.1 Optical Lattices

The creation of optical lattices takes place through laser beam interference that produces periodic potential spaces for trapping neutral atoms. The experimental arrangements facilitate the confinement of ultracold atoms into specific locations that analogously represent the crystal arrangement of electrons. Researchers maintain extensive control over the operating parameters including the depth of the lattice and atomic interaction forces and spatial design dimensions in optical lattices [1]. The ability to change parameters enables optical lattices to serve as a strong platform for simulating and examining condensed matter physics models particularly the Bose-Hubbard model [2].

1.2 Superfluid–Mott Insulator Transition

The Bose-Hubbard model is a theoretical model that tells about the interaction between the bosons trapped in an optical lattice. It is a mathematical framework that has parameters like the hopping term J , On-site interaction U , and chemical potential μ . The competition between the parameters like hopping term and on-site interaction term decides the phase whether it will be Mott-Insulator or

Superfluid Phase. [3, 4].

When U starts dominating J , the atoms are localized, leading to the Mott insulator phase, where each site has a fixed number of atoms and there is no global phase coherence. On the other hand, when J dominates, atoms delocalize across the lattice, resulting in the superfluid phase where long-range phase coherence appears. This transition between the Mott insulator and the superfluid is a quantum phase transition, driven purely by quantum fluctuations at zero temperature [5].

1.3 Literature Review

Scientific investigations and experimental work have intensely researched quantum phase transitions in systems described by the Bose-Hubbard model. A transition between Mott insulator (MI) and superfluid (SF) phases happens through quantum phase when the system controls the hopping amplitude J to balance between kinetic energy and interaction energy. Recently research has concentrated on understanding how systems transform during these transitions when controlling parameters at finite rates particularly during such transformations.

A fundamental theoretical tool to describe these system behaviors is the Kibble–Zurek (KZ) mechanism which researchers use in references [6, 7]. During slow system-wide phase transition movements the divergent relaxation times near critical points will make the system lose equilibrium status. Different portions of the system fail to match their phase selection thus producing defects or domains during this process. The model defines that defect densities throughout the system are directly proportional to quench speed rate under a generic mathematical power relation.

$$n_{\text{defects}} \propto \tau_Q^{-\frac{d\nu}{1+z\nu}} \quad (1.1)$$

The density of topological defects depends on several factors through the rela-

relationship $n_{\text{defects}} \propto \tau_Q^{-\frac{d\nu}{1+z\nu}}$. This expression includes three key variables: topological defect density (n_{defects}), quench time (τ_Q) and system dimensionalities d together with ν being the critical exponent for the correlation length and z defined as the dynamic critical exponent.

Timescale analysis of the two-dimensional Bose-Hubbard model occurred through Shimizu et al. [8] who utilized time-dependent Gutzwiller mean-field simulations. The Mott insulator phase features small values of superfluid order parameter ψ until J exceeds its critical threshold at which growth finally appears. Shimizu et al.'s research revealed the system operates on two separate time-scales during their investigation: first the domain freezes during critical slowing down at the freeze-out time $\hat{t}$ followed by the formation of larger coherent structures during the coarsening period at time t_{eq} . The research team found evidence of KZ predictions but observed oscillatory behaviors together with an extended domain growth past critical threshold in experimental data.

Additional confirming results emerged from the study Braun et al. conducted in [9]. Scientists monitored coherence development through optical lattices containing ultracold atoms when performing SF-MI transition quenches at slow rates. Experimental tests found that the duration of the quench process produced a direct proportional relationship with final coherence length. The experimental data showed changing critical exponents because both forcing protocols alongside interparticle coupling strength regulations led to these value changes but KZ scaling was still valid.

Kaur et al. conducted a research analysis which focused on two-component Bose mixtures in their article [10]. Critical slowing down exhibits unique behavior during first-order phase transitions because the research established that multiple-component systems produce unconventional dynamical patterns compared to single-component systems. The Kibble-Zurek principles remain true but the specific outcomes depend heavily on system characteristics together with the transition mechanism.

Realistic system quantum phase transition dynamics show modified domains

and emerging delays and system-size-dependent features without affecting the initial value predictions from the Kibble–Zurek theory model. The findings of research permit scientists to execute theoretical and numerical assessments that advance our understanding of the quantum dynamics at various phase transition stages.

1.4 Dynamics: A Brief Review

Kibble-Zurek Mechanism states that if the quantum system is in equilibrium and if we start changing the parameters of the system like hopping amplitude J as a function of time t linearly at a very slow rate near the critical point the system enters into a non-equilibrium state. There is a critical slowing down of the system near the critical point where the relaxation time which is the time required for the system to adjust to changing parameters becomes very large and the system falls out of equilibrium and forms topological defects in the new phase such as vortices, domain walls etc. [7]

Studies have verified this behavior experimentally. They have observed how the dynamics depend on quench rate and system parameters. For example, Braun et al. [9] observed perfect agreement between experimental observations and numerical simulations for the resulting coherence length. Also for intermediate quench velocities, a power-law behavior of the coherence length was observed which resembled the Kibble-Zurek scaling law. Similarly, Shimizu et al. [8] performed dynamics on a 2D Bose-Hubbard model which was solved using the time-dependent Gutzwiller method. They have observed oscillations formed in the time evolution of order parameters and also the formation of defects as the system is linearly quenched.

1.5 Motivation

In modern physics understanding the time evolution of the quantum phase transition in many body problems is very challenging. Due to the advancements

of the Ultracold experiments, it is now possible to explore, study, and experiment with quantum phase transitions happening in the Standard Bose-Hubbard Model [1, 5]. When the quantum system is in the equilibrium phase it is quite simple to study than when the system falls out of equilibrium when it is linearly quenched. Due to this, the system becomes much more complex to understand leading to non-equilibrium effects such as defect formation and domain growth, which are described by the Kibble–Zurek mechanism [6, 7]. Recently a theoretical study conducted by Shimizu et al. [8], has shown that as the quenching is done slowly across the MI-SF transition, interesting dynamical features such as oscillations in the order parameter, and freeze-out time were observed. This helped to verify the Kibble-Zurek Mechanism and also provided a deeper insight into the non-equilibrium behavior of the system. In this project, I aim to study the dynamics of the two-dimensional Bose-Hubbard model during a linear quench, tracking the evolution of the superfluid order parameter, and analyzing the formation of domains as the system evolves from the Mott insulator to the superfluid phase.

1.6 Aim

To study the dynamics of Quantum Phase Transition from Mott Insulator to Superfluid in the 2D Bose Hubbard Model.

1.7 Plan of the Report

The structure of this report is organized as follows:

- **Chapter 1** introduced the theory for understanding the fundamental concepts like Optical Lattice, SF-MI Transition. It also included Literature Review, Motivation for doing this project and the Objective of the project.
- **Chapter 2** explains the Methodology used to solve the model, numerical method used to conduct the time evolution dynamics, studying the

methods to extract freeze-out time and calculating domain size.

- **Chapter 3** will consist of the results obtained from the simulations. Also the interpretation of those Results. It includes the static phase diagram, graph of the time evolution of the order parameter during the quench and the evolution of domain sizes. The Analysis of Freeze-Out dynamics across the quench is also shown. Snapshots of spatial variation of ψ is also included.
- **Chapter 4** is the conclusion of the report which summarizes everything like the key findings and interpretations. Also suggesting the future prospects.

Chapter 2

Methodology

2.1 Mean Field Theory on 2D Bose-Hubbard Model

The standard Bose-Hubbard model is one of the simplest and most important theoretical models used to describe bosonic particles (like atoms) on an optical lattice. It is especially useful in ultra-cold atom systems, where atoms are trapped in optical lattices created using laser light. The model helps us to understand how particles behave when they hop between the neighboring lattice sites and interact with each other at the same site.

The Hamiltonian of the Bose-Hubbard model is:

$$\hat{H} = -J \sum_{\langle i,j \rangle} (\hat{a}_i^\dagger \hat{a}_j + \text{h.c.}) + \frac{U}{2} \sum_i \hat{n}_i(\hat{n}_i - 1) - \mu \sum_i \hat{n}_i \quad (2.1)$$

where

- J is the **hopping amplitude** that allows bosons to hop or move between neighboring sites i and j ,
- U is the **on-site interaction strength**, representing repulsion between bosons occupying the same site,

- μ is the **chemical potential**, controlling the number of particles,
- $\hat{a}_i^\dagger$ and $\hat{a}_i$ are the **bosonic creation and annihilation operators** at site i ,
- $\hat{n}_i = \hat{a}_i^\dagger \hat{a}_i$ is the **number operator** that counts how many particles are on site i .

For large systems the two-dimensional Bose-Hubbard model is very challenging and complex to solve exactly. Therefore we apply **Mean Field Theory (MFT)** to solve this two-dimensional Hamiltonian to simplify problem by neglecting fluctuations and making a useful approximation.

The Mean Field Approximation

The basic strategy of MFT is to focus on a single site and assuming an **average effect** from the neighboring sites rather than accounting for full quantum correlations between them. The procedure also involves ignoring the fluctuations.

Hence we use Mean Field Approximations for the hopping term:

$$a_i = \langle a_i \rangle + \delta a_i \quad (2.2)$$

$$\langle \hat{a}_i \rangle = \psi_i \quad (2.3)$$

$$a_i = \psi_i + \delta a_i \quad (2.4)$$

Here, $\psi_i = \psi e^{i\phi}$ is the complex superfluid order parameter. This parameter ψ_i is the average value of the annihilation operator which tells us if the system is in the **superfluid phase** ($\psi \neq 0$) or the **Mott insulating phase** ($\psi = 0$).

We then make a decoupling approximation for the hopping term:

$$\hat{a}_i^\dagger \hat{a}_j + \hat{a}_j^\dagger \hat{a}_i \approx ((a_i^\dagger \psi_j + a_i \psi_j^*) - \frac{(\psi_i^* \psi_j + \psi_i \psi_j^*)}{2}) \quad (2.5)$$

Substituting in eq.2.1 the original many-body Hamiltonian is now turned

into an effective **single-site Hamiltonian** for each lattice site:

$$\hat{H}_{\text{MF}}^{(i)} = -J \sum_{j=1}^z ((a_i^\dagger \psi_j + a_i \psi_j^*) - \frac{(\psi_i^* \psi_j + \psi_i \psi_j^*)}{2}) + \sum_i \frac{U}{2} n_i (n_i - 1) - \sum_i \mu n_i \quad (2.6)$$

Here, z is the **coordination number** (number of neighbors each site has), which is $z = 4$ for a 2D square lattice.

Self-Consistent Calculation

To find the correct value of ψ_i , we solve the effective single-site Hamiltonian numerically. The steps are:

1. Guess an initial value for ψ_i .
2. Diagonalize $\hat{H}_{\text{MF}}^{(i)}$ to find the ground state.
3. Calculate a new value of $\psi_i = \langle \hat{a}_i \rangle$ from the ground state.
4. Repeat steps 2-3 until ψ converges (i.e., doesn't change much between iterations).

The value of the superfluid order parameter as a function of J/U and μ/U is given by the above **self-consistent procedure**. The phase diagram which is obtained by using this methods shows us the regions where the system could be in Superfluid Phase or Mott-Insulating Phase.

Why Use Mean Field Theory?

Although MFT Suppresses quantum fluctuations between different sites, it still captures the main physics of the Bose-Hubbard model, especially the phase transition between the superfluid and Mott insulator. It's a useful method for studying:

- The rough **critical point** where the phase transition occurs,
- How the **order parameter** changes across the transition,

- The **dynamics** during slow changes (or “quenches”) of the system parameters, as done in many quantum simulation experiments.

In particular, for large 2D systems, MFT is a good starting point to simulate the response of the system to changes in hopping J , like in quench protocols where J is varied slowly in time across the phase transition.

2.2 Dynamics of the Quench Process

In order to study the real-time dynamics of the quantum phase transition in the two-dimensional Bose-Hubbard model, the system was first prepared in its ground state corresponding to the Mott insulating phase. The simulation was carried out on a 70×70 square lattice with periodic boundary conditions (PBC) applied along both directions to minimize edge effects. The value of on-site interaction strength was fixed at $U = 1.0$, and the total quench time was set to $\tau_Q = 50$ (in units of $\hbar/U$).

The ground state was obtained by minimizing the order parameter $\psi = \langle \hat{a}_i \rangle$ at a low hopping amplitude $J_i = 0.03$, where the system is deep in the Mott regime. In the Mott phase, this quantity is close to zero since particles are localized and there is no long-range phase coherence.

Once the initial ground state $|\text{GS}(t)\rangle$ was obtained by applying the Self Consistent method, the system was evolved in time according to the time-dependent Schrödinger equation:

$$\frac{d}{dt}|\text{GS}(t)\rangle = -\frac{i}{\hbar}H(t)|\text{GS}(t)\rangle.$$

where $\hbar$ was taken to be equal to 1. At each iteration or time step, the hopping amplitude J was updated according to the quench protocol, and the corresponding ground state vector $|\text{GS}(t)\rangle$ was used in order to calculate the correspond Superfluid order parameter.

The time evolution was performed numerically by using the fifth-order Runge-Kutta (RK5) method. It was found to be computational efficient and accurate

than other methods like Euler's method or second order Runge Kutta method. After applying RK5 method the wavefunction was being updated at each iteration, which allowed us to track how the quantum state of the system changed as external parameters (like the hopping amplitude) were varied.

The Quantities calculated at each iteration included:

- $\langle \hat{a}_i \rangle = \psi_i$ - the local superfluid order parameter,
- $\langle \hat{n}_i \rangle = \rho_i$ - the local particle density.

We calculated the modulus of the Complex Superfluid order parameter $|\psi_i|$, summed it over all sites, and divided it by the total number of sites to obtain the average of the superfluid order parameter i.e. $|\psi_{avg}|$. Similarly we calculated particle density ρ_{avg}

These quantities were calculated for every value of $J(t)$ and accordingly recorded. The trend of how the time evolution of Superfluid order and particle density takes place was observed by plotting the graph of $|\psi_{avg}|, \rho_{avg}$ vs. $J(t)$.

To drive the phase transition, we used a *linear quench protocol*, where the hopping amplitude J was increased smoothly in time according to:

$$J(t) = J_i + \frac{(J_c - J_i)}{T_Q}(t + T_Q) \quad (2.7)$$

where

- $J_i = 0.03$ is the initial hopping amplitude,
- $J_c = 0.042$ is the critical hopping amplitude,
- T_Q is the total quench duration ($\tau_Q = 50$),
- The final hopping value after the quench is $J_f = 0.1$,
- t runs from t_i to t_f , where $t_i = -\tau_Q$ and $t_f = \tau_Q(-1.0 + (\frac{J_f - J_i}{J_c - J_i}))$
- The time step for evolution was $\Delta t = 0.0005833 (\hbar/U)$.

This linear ramp was applied symmetrically over $t \in [t_i, t_f]$.

The time factor introduced so far in our simulations is of the units of $\hbar/U$ where $\hbar = 1$ and $U = 1$. Hence the physical time which is in units of seconds will be equal to:

$$t_{\text{phys}} = t \frac{\hbar}{U} \quad \text{seconds}$$

Therefore one “tick” of the dimensionless time variable $t = 1$ corresponds to

$$t_{\text{phys}} = \frac{\hbar}{U} \quad \text{seconds.}$$

2.3 Calculation of Freeze-Out Time Using the Overlap Function

For the freeze-out time t_{freeze} to be determined during the different quench dynamics, the overlap function between the initial ground state and the time-evolved state was used.

The method was conducted in the following way:

1. **Initial State Preparation:** By doing the self-consistency method we minimized the energy of the system and achieved the ground state $\text{GS}(0)$ corresponding to the initial hopping amplitude $J_i = 0.03$.
2. **Overlap Function $O(t)$:** After applying the Time evolution on the ground state we used the respective ground states belonging to that particular $J(t)$ and calculated the overlap function $O(t)$ by finding inner product between the time-evolved state and the instantaneous ground state.

$$O(t) = |\langle \text{GS}(0) | \text{GS}(t) \rangle|, \quad (2.8)$$

where $\text{GS}(t)$ is the wavefunction evolved under the time-dependent Hamiltonian and $\text{GS}(0)$ is the instantaneous ground state.

3. **Detection of Freeze-Out Time:** At the start of the process $O(t)$ stays

near 1 representing adiabatic evolution. The approach to the critical point causes $O(t)$ values to separate from 1. Scientists defined the freeze-out time t_{freeze} through its detection at the very first instance when

$$|O(t) - 1| > 10^{-2}.$$

A programming code detected the first deviation occurrence where it extracted both τ_Q and t_{freeze} measurements.

This process helped to determine freeze-out times across various quench times which allowed an investigation of t_{freeze} dependence on τ_Q .

2.4 Analysis of Domain Size

The research examined both the average superfluid order parameter development as well as the growth pattern of superfluid regions. The site count having values of $\psi > 0.1$ throughout the lattice at different stages during evolution was computed. Each row was examined to identify the one with the highest number of such superfluid sites which generated a corresponding maximum count recorded against $J(t)$.

During the quench process we measured the maximum domain size to track the coherent superfluid regions that developed from the Mott insulating phase to the superfluid phase.

Chapter 3

Results

3.1 Phase transition

The Mott insulating (MI) phase and superfluid (SF) phases exist alongside each other in the phase boundary area which can be observed in graph 3.1. The evaluation determined that a hopping amplitude J_c equaled 4.2×10^{-2} when the interaction strength U remained at 1.0. The Figure 3.1 provides direct visual evidence for how the Mott insulating phase transforms into a superfluid phase. Each site of the MI phase exhibits localized particles because of strong interparticle repulsions leading to fixed particle counts and zero superfluid order. The phase transition in the system occurs at each lobe where Mott insulating behavior exists with specific integer boson filling on each site including one boson per site in the first lobe and two bosons per site in the second lobe. The system achieves superfluidity when hopping amplitude J exceeds its critical value because particle delocalization and phase coherence across the lattice occur due to kinetic energy surpassing interaction strength.

Mean field theory has theoretically determined the critical SF–MI transition point at unit filling within two-dimensional systems through an analysis for which $J/U = 1/(z \times 5.8)$ matches the coordination number z of the lattice. The theoretical model indicates that for the square lattice with $z = 4$ the ratio J/U

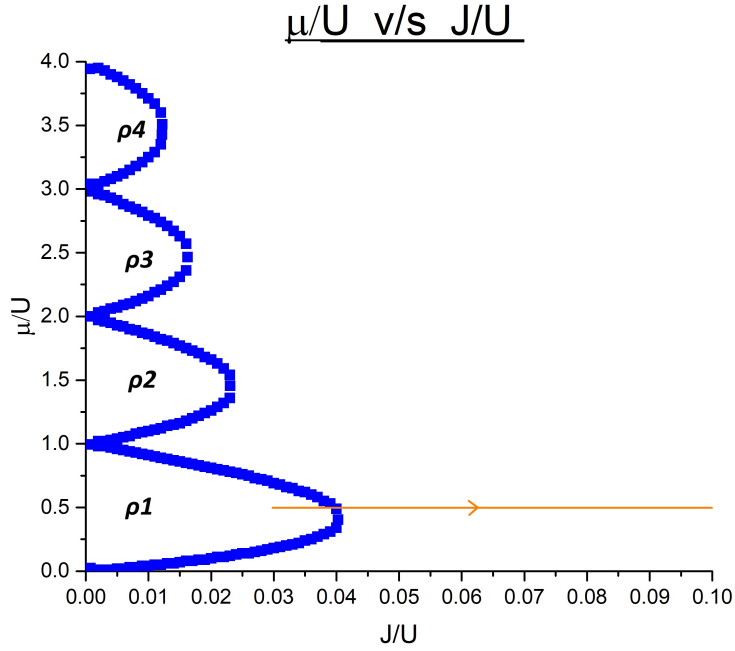


Figure 3.1: Phase diagram showing the Mott insulator to superfluid transition for $U = 1.0$. The critical point is observed at $J_c = 0.042$, in agreement with theoretical predictions. The orange line shows where the time evolution will be conducted by quenching $J(t)$ linearly at $\mu = 0.5$.

becomes approximately 0.043. This value aligns with the observed critical point in the phase diagram. The presented magnitude of hopping strength matches the critical strength value documented in Ref. [8].

The orange line shows where the time evolution will be conducted by applying a linear quench of at $\mu = 0.5$.

3.2 Linear Quench Profile: Variation of Hopping Amplitude $J(t)$ with time t .

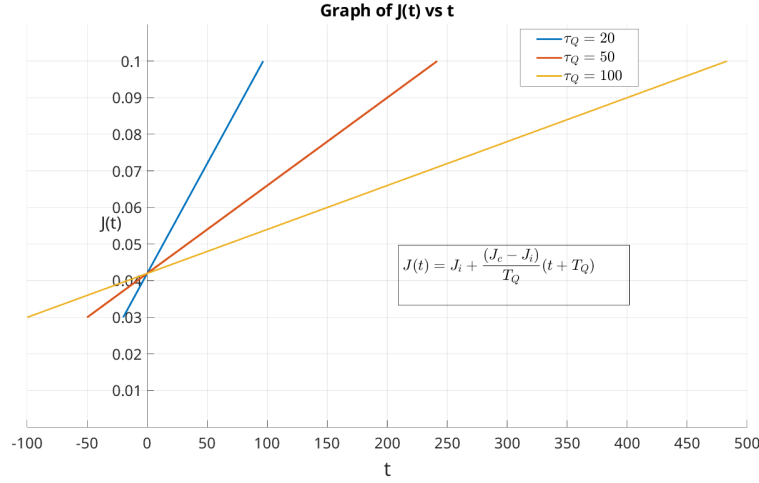


Figure 3.2: Variation of the dimensionless hopping amplitude $J(t)/U$ as a function of the dimensionless time t (in units of $\hbar/U$, with $\hbar = 1$, $U = 1$) for quench durations $\tau_Q = 20, 50, 100$ (also in units of $\hbar/U$).

The hopping amplitude varies linearly as time t increases. The blue line is for linear quench profile having $\tau_Q = 20$, Similarly red is for $\tau_Q = 50$ and yellow is for $\tau_Q = 100$. Here the $J(t)$ was increased in accordance with time show in eq.2.7 where $J_i = 0.03$, $J_c = 0.042$ and respective τ_Q values. The time factor t ran from $t_i = -\tau_Q$ to $t_f = \tau_Q(-1.0 + (\frac{J_f - J_i}{J_c - J_i}))$.

The Quench rate which tells us how fast the parameter is changed is inversely proportional to the Quench time τ_Q . As can be seen from the graph for higher values of quench time the Quench rate will be less, and for lower values of quench time the Quench rate will be faster.

Next, we will see the results of linear quench applied to the Mott Insulator phase at value $\tau_Q = 50$.

3.3 Time Evolution

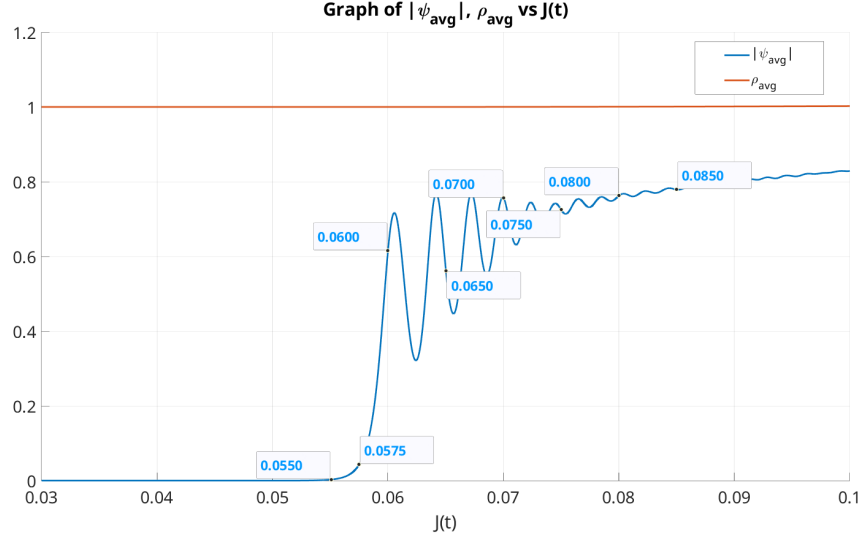


Figure 3.3: Variation of the average order parameter ψ_{avg} and ρ_{avg} as a function of the hopping amplitude $J(t)$. Initially, ψ_{avg} is small, indicating a Mott insulating phase. As $J(t)$ increases, ψ_{avg} rises, signaling the onset of superfluid domains. The values taken were $\tau_Q = 50$, $J_i = 0.03$, $J_f = 0.10$ and $J_c = 0.042$.

The time evolution graph of the average superfluid order parameter ψ_{avg} and the average particle density ρ_{avg} provides information of how the system transitions from the Mott insulating (MI) phase to the superfluid (SF) phase during the linear quench.

Initially, when the hopping amplitude $J(t)$ is small, the system is deep in the Mott insulating regime. In this phase, strong on-site interactions dominate, and particles are localized at individual lattice sites. As a result, the superfluid order parameter ψ_{avg} remains close to zero, indicating that the particles are localized and there is no phase coherence between them. In other words, the system is in a Mott insulator state.

The ρ_{avg} remains almost constant throughout the quench process. This reflects particle number conservation during the dynamics.

As $J(t)$ is gradually increased during the quench, the competition between

kinetic energy (hopping) $J(t)$ and interaction energy U shifts in favor of hopping. When $J(t)$ reaches a specific threshold close to the critical point ψ_{avg} experiences a rapid increase. The sudden rise establishes the emergence of superfluid domains as particles transition to a delocalized state with developing long-range phase coherence. The average local particle number remains stable throughout this phase transition period as indicated by constant ρ_{avg} values.

During the growth of ψ_{avg} we notice minimal variations in the data. These oscillations match predictions of the Kibble–Zurek mechanism concerning finite-time continuous phase transitions which cause domains and defects from critical slowing effects [6, 8].

Based on the Kibble–Zurek (KZ) mechanism it becomes apparent that near the critical point the system develops an infinite relaxation time thus preventing immediate adaptation to parameter changes. As a result, different regions of the system fall out of equilibrium and choose independent phases, leading to the formation of domains and defects. The subsequent oscillations in ψ_{avg} reflect the coarsening dynamics, where domains grow and try to establish a uniform phase across the system.

Thus, the behavior observed in the ψ_{avg} and ρ_{avg} vs $J(t)$ graph not only confirms the expected Mott insulator to superfluid transition but also provides evidence for nonequilibrium effects such as defect formation, delayed establishment of global coherence, and conservation of particle number throughout the quench. The following snapshots correspond to the spatial distribution of ψ at selected $J(t)$ values during the time evolution.

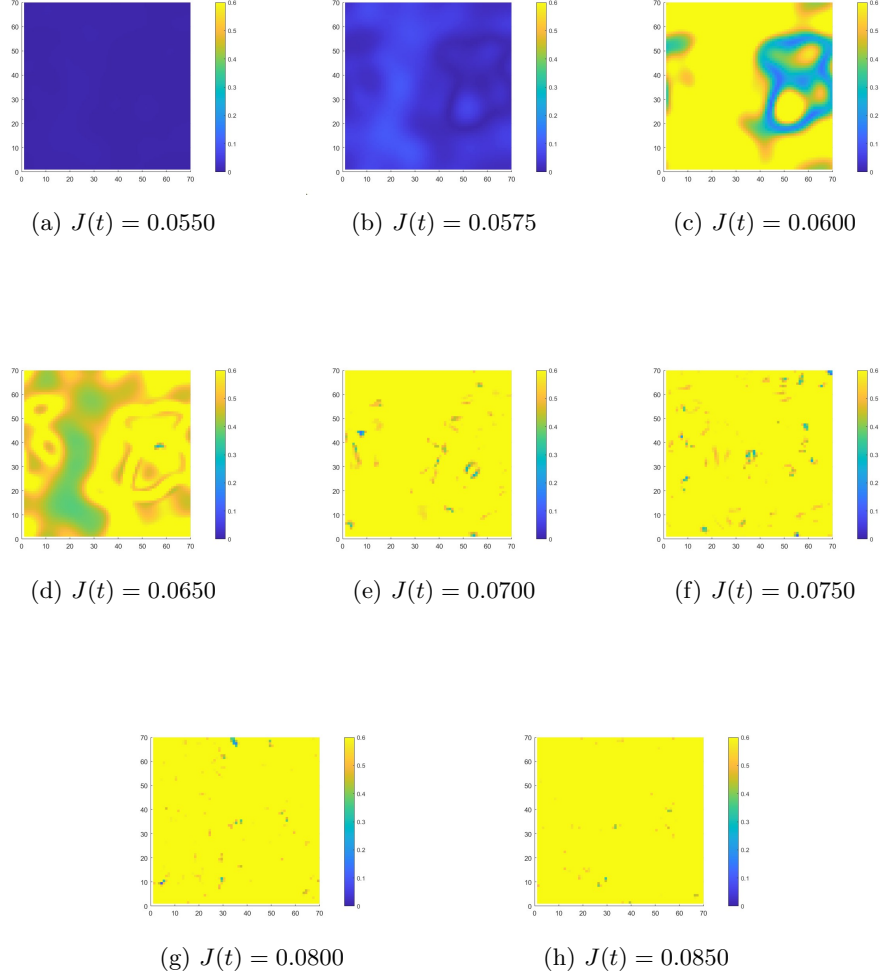


Figure 3.4: Snapshots showing the spatial distribution of $|\psi|$ during the quench. As $J(t)$ is increased, superfluid order parameter emerge and grow which indicates the development of long-range phase coherence. On x and y axis is total no. of sites i.e. 70X70 sites.

Initially, when the hopping amplitude $J(t)$ is small, the average superfluid order parameter ψ_{avg} remains close to zero. This indicates that the particles are localized on their respective lattice sites without any phase coherence between them, characteristic of the Mott insulating phase.

As $J(t)$ is increased during the quench, ψ_{avg} begins to rise, signaling the onset of phase coherence and the transition towards a superfluid phase. This evolution can also be clearly seen in the snapshots shown in Fig. 3.4.

In Fig. 3.4(a), corresponding to $J(t) = 0.0550$, we observe that ψ is still mostly small across the system, with only a few localized regions showing slightly higher values. This suggests that the system remains largely in the Mott insulating regime.

In Fig. 3.4(b), at $J(t) = 0.0575$, small isolated superfluid patches start to appear but remain disconnected.

Moving to Fig. 3.4(c) and Fig. 3.4(d), corresponding to $J(t) = 0.0600$ and 0.0650 , respectively, we observe that these superfluid patches grow larger and begin to merge, indicating the formation of connected superfluid domains.

The trend continues in Fig. 3.4(e) and Fig. 3.4(f), where at higher $J(t)$ values such as 0.0700 and 0.0750 , the superfluid regions become significantly larger and more continuous.

Finally, in Fig. 3.4(g) and Fig. 3.4(h), at $J(t) = 0.0800$ and 0.0850 , the system exhibits strong, nearly global phase coherence, indicating that it has transitioned deep into the superfluid phase.

Thus, by following the evolution of the snapshots across increasing $J(t)$, we can visually observe how the system gradually develops coherence, confirming the physical picture obtained from the ψ_{avg} versus $J(t)$ graph.

3.4 Domain Size Growth During the Quench

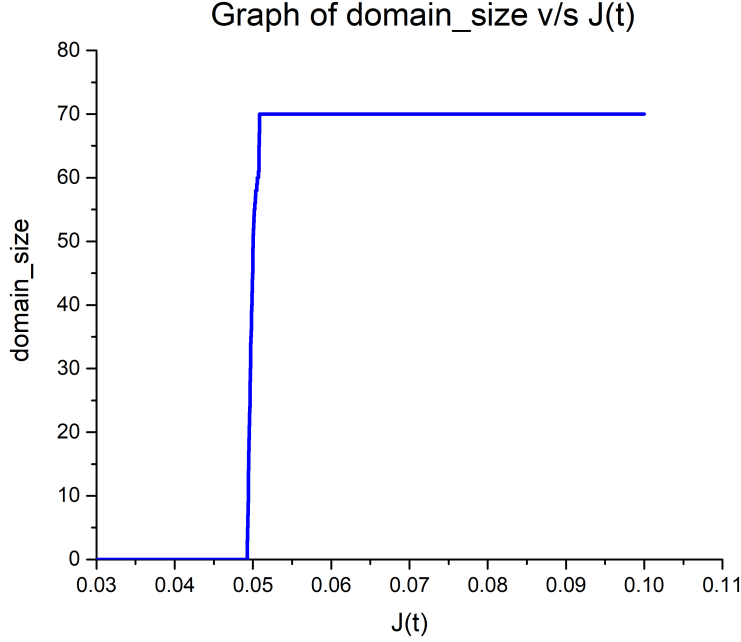


Figure 3.5: The maximum number of superfluid sites exceeding a threshold value $\psi > 0.1$ develops across a row under variations of hopping amplitude $J(t)$ according to the plot in Fig.3.3 A distinct increase of MaxCount occurs near the critical region before reaching saturation at higher $J(t)$ values since superfluid domains spread throughout space. The simulation used values of $\tau_Q = 50$ alongside $J_i = 0.03, J_f = 0.10$ while $J_c = 0.042$.

Here domain size tells us the maximum number of lattice sites in a row where the superfluid condition $\psi > 0.1$ is satisfied. As we can see from the above graph, initially, the maximum count is very small when the system is deep in the Mott insulating phase starting from $J_i = 0.03$.

The MaxCount measure grows steadily during the time evolution because superfluid domains both materialize and spread throughout the lattice system. The system exhibits a dramatic boost in MaxCount when it reaches $J(t) = 0.049$ which indicates the superfluid domains start increasing as the system approaches

the theoretical critical point.

The MaxCount achieves full capacity matching the number of sites in a row when the value of $J(t)$ becomes very high which demonstrates that atoms develop percolation capability throughout the entire lattice. The achievement of maximum MaxCount value leads to spatial superfluid distribution but does not confirm complete system-wide phase coherence. The time evolution graph shows temporal oscillations in the average order parameter $\langle\psi\rangle$ which indicates phase coherence establishment continues beyond when the superfluid spreads across the system spatially.

3.5 Analysis of Freeze-Out Dynamics Across the Quench

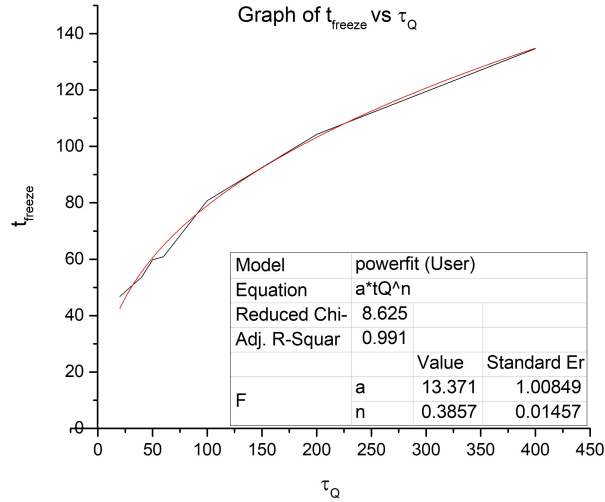


Figure 3.6: Variation of freeze-out time $\hat{t} = t_{\text{freeze}}$ as a function of quench time τ_Q . The data points are power-fitted, and the obtained constant is approximately 0.38, close to the reported value of 0.36 in Ref. [8].

The above plot shows that as the freeze-out time t_{freeze} increases with the quench time τ_Q according to the power-law scaling, aligning with the predictions of the Kibble–Zurek (KZ) mechanism.

Later, a log-log plot of t_{freeze} vs. τ_Q is performed to linearly fit it and extract the exponents confirming that the slope approximately matches the desired results.

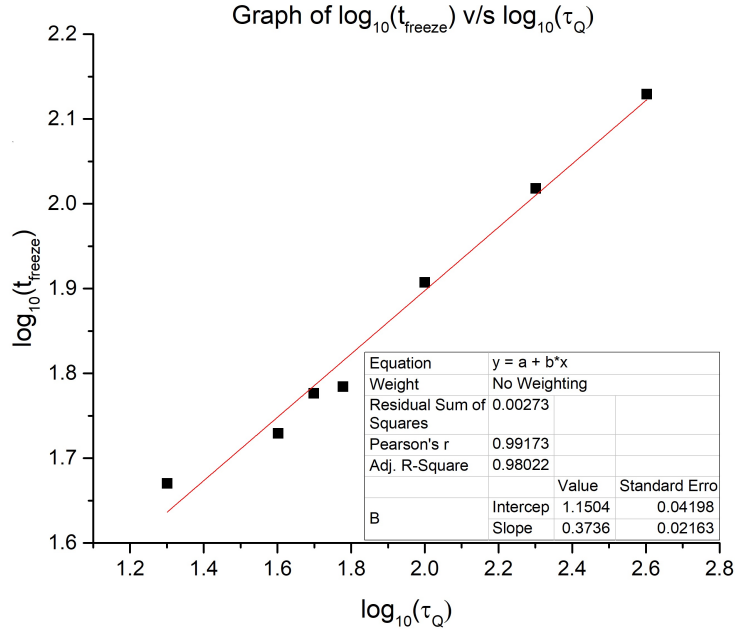


Figure 3.7: Plot of t_{freeze} vs τ_Q on the logarithmic scale. The data points are fitted to a linear relation, and the obtained constant is approximately 0.37, close to the reported value of 0.36 in Ref. [8].

As we can see from the above graph, the logarithmic plot of freeze-out time t_{freeze} increases linearly with the quench time τ_Q , consistent with the Kibble–Zurek (KZ) mechanism. A linear fit to the t_{freeze} vs τ_Q data yields a slope of approximately 0.37. This value is consistent with the value 0.36 reported by Shimizu et al. [8]. Overall, the result supports the Kibble–Zurek scaling behavior.

Chapter 4

Conclusion

In this project, the mean-field approximation was used to solve the Hamiltonian two-dimensional Bose-Hubbard model. We diagonalized the Hamiltonian, achieved the ground state, and computed the Superfluid order parameter and particle density using the ground state.

A varied the hopping amplitude linearly to quench the system to evolve it across the Mott insulator-superfluid phase transition after it had been prepared in the Mott insulating ground state for the simulation.

After applying time evolution we observed the transition from the localized Mott insulator Phase to a delocalized superfluid Phase where we saw that the average order parameter $|\psi_{avg}|$, rose from near zero and began to exhibit oscillations near the critical point as the system entered the superfluid regime and beyond the critical point, as $J(t)$ increased further, the oscillations stabilized and the superfluid order parameter started settling to a nearly constant value.

In the domain size evolution we observed that as we increased the hopping strength, the little islands of superfluid first appeared and subsequently expanded to occupy entire rows. We observed the order parameter fluctuating until it eventually settled since the system wasn't yet completely in sync, even when every site in a row was "superfluid."

We also verified the power-law scaling of freeze-out time t_{freeze} vs. the quench

time τ_Q , aligning with the predictions of the Kibble–Zurek (KZ) mechanism. Future work could involve finding the critical exponents. Applying the same procedure of Time Evolution to the Extended Bose-Hubbard model and studying its dynamics for various Quantum Phase Transitions.

Bibliography

- [1] Immanuel Bloch. Ultracold quantum gases in optical lattices. *Nature Physics*, 1(1):23–30, 2005.
- [2] Maciej Lewenstein, Anna Sanpera, Veronica Ahufinger, Bogdan Damski, Aditi Sen, and Ujjwal Sen. Ultracold atoms in optical lattices: mimicking condensed matter physics and beyond. *Advances in Physics*, 56(2):243–379, 2007.
- [3] Matthew P A Fisher, Peter B Weichman, G Grinstein, and Daniel S Fisher. Boson localization and the superfluid-insulator transition. *Physical Review B*, 40(1):546, 1989.
- [4] D Jaksch, C Bruder, J I Cirac, C W Gardiner, and P Zoller. Cold bosonic atoms in optical lattices. *Physical Review Letters*, 81(15):3108, 1998.
- [5] Markus Greiner, Olaf Mandel, Tilman Esslinger, Theodor W Hänsch, and Immanuel Bloch. Quantum phase transition from a superfluid to a mott insulator in a gas of ultracold atoms. *Nature*, 415(6867):39–44, 2002.
- [6] Wojciech H Zurek. Dynamics of topological defects and phase transitions. *Physics Reports*, 276(4-6):177–221, 1996.
- [7] Anatoli Polkovnikov, Krishnendu Sengupta, Alessandro Silva, and Mukund Vengalattore. Colloquium: Nonequilibrium dynamics of closed interacting quantum systems. *Reviews of Modern Physics*, 83(3):863, 2011.

- [8] Yuki Shimizu, Takeshi Kano, and Ippei Danshita. Dynamics of a quantum phase transition in the bose–hubbard model: Kibble–zurek mechanism and beyond. *Physical Review A*, 97(3):033626, 2018.
- [9] S. Braun, M. Friesdorf, S. S. Hodgman, M. Schreiber, J. P. Ronzheimer, A. Riera, L. del Rio, I. Bloch, J. Eisert, and U. Schneider. Emergence of coherence and the dynamics of quantum phase transitions. *Proceedings of the National Academy of Sciences*, 112(12):3641–3646, 2015.
- [10] Pardeep Kaur, Kuldeep Suthar, Dilip Angom, and Sandeep Gautam. Out-of-equilibrium dynamics of bose-bose mixtures in optical lattices. *Physical Review A*, 109(1):013308, 2024.